

Multiplying polynomials



1

a

i y^2

ii $3y$

iii $7y$

iv 21

b $y^2 + 10y + 21$

2

a $ab - 6a + 9b - 54$

b $28rs + 8r + 49s + 14$

c $x^2 + 7x + 10$

d $x^2 + 12x + 35$

e $v^2 + 18v + 80$

f $x^2 + 3x - 10$

g $x^2 - 5x + 6$

h $m^2 - 3m - 18$

i $5m^2 - 7m - 24$

j $35w^2 + 39w + 10$

k $42y^2 + 6y - 36$

l $4c^2 - 56.2c + 2.8$

m $\frac{1}{9}y^2 + \frac{1}{36}y - \frac{1}{24}$

n $-90y^2 + 26y + 48$

o $9y^2 + 90y + 144$

p $7x^2 - 42x - 189$

q $-x^2 - 7x - 10$

r $-5y^2 - 45y - 100$

s $-4y^2 + 12y + 112$

t $-75x^2 + 110x - 40$

u $6x^2 - 12x - 18$

v $6x^{10} + 13x^5y^4 + 5y^8$

w $-30x^2 - 42x + 34$

x $-8x^2 + 40x + 72$

y $18x^2 - \frac{69x}{2} - 8$

z $-4x^2 - 6x + 32$

3 $(12x^2 + 144x + 384) \text{ m}^3$

4 $42x^2 + 59xy + 20y^2$

5

a $(h + 9) \text{ cm}$

b $(w + 8) \text{ cm}$

c $\left(4h + \frac{9w}{2} + 36\right) \text{ cm}^2$

6 $(128y^2 + 48y) \text{ cm}^3$

7

a $5a^3 + 8a^2 - 2a + 4$

c $-25a^3 - 50a^2 - 45a - 20$

e $6a^3 - 13a^2b - 2ab^2 + 12b^3$

g $-56x^2 + 24xy - 12x + 36y + 108$



b $u^3 + u^2 - 15u - 20$

d $20c^4 - 15c^3 + 2c^2 + 6c - 4$

f $-12m^3 - 20m^2 + 8$

8 True

9 -13

10

a 12

b -18

c Yes

11

a 4

b 5

12 $12t^7$

13

a Yes

b No

c No

d No

14

a No

b Yes

c Yes

d No

e Yes

f No

15

a Yes

b No

c Yes

d No

16

a No

b Yes

c Yes

d Yes

17 $(8y^3 + 22y^2 + 17y + 3) \text{ ft}^3$

18

a $(10 - 2x) \text{ cm}$

b $(6 - 2x) \text{ cm}$

c $(4x^3 - 32x^2 + 60x) \text{ cm}^3$



Additional questions

19

- a 4 and -28
- b 1 and -11
- c Answers may vary. 2, 6, and 18 is one possibility. 3, 4, and 18 is another.

20

- a The new dimensions are $(5 + 2x)$ and $(7 + 2x)$
- b $4x^2 + 24x + 35$
- c $4x + 24$

21

- a The new dimensions are $(12 - 2x)$, $(12 - 2x)$, and $(5.5 - 2x)$
- b Jerome will have a volume of $-4x^3 + 59x^2 - 276x + 396$ left to fill.

22

- a $(10x)(10x) = 100x^2$, but Anette put $100x$ instead. Her final answer should have been $100x^2 + 20x + 30x + 6 = 100x^2 + 50x + 6$
- b

	x^2	$10x$	2
10x	$10x^3$	$100x^2$	$20x$
3	$3x^2$	$30x$	6

- c $10x^3 + 103x^2 + 50x + 6$

23 The result will be a polynomial with degree 7.

24



- a Agree. The maximum number of terms that can be produced by multiplying two binomials is $2 \times 2 = 4$ if no terms combine.
- b Answers will vary. A monomial multiplied by a five term polynomial will always results in a five term polynomial. Another possible solution is a binomial multiplied by a trinomial that results in a 4th degree polynomial with exactly one pair of terms combining.
For example, $(x^2 + 3)(x^2 - 2x + 1) = x^4 - 2x^3 + 2x^2 - 6x + 3$